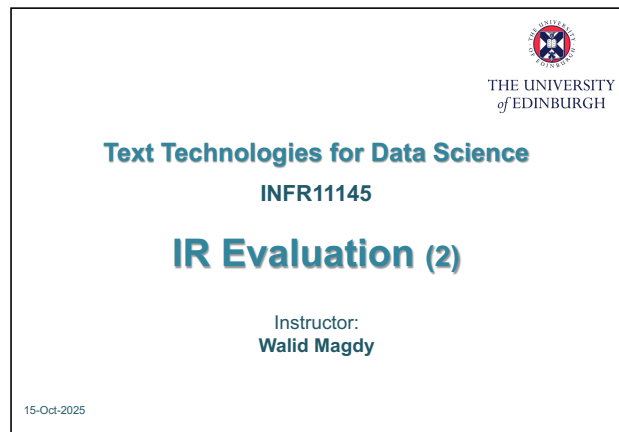


IR Evaluation 2: Test Collections、Pooling 与显著性检验

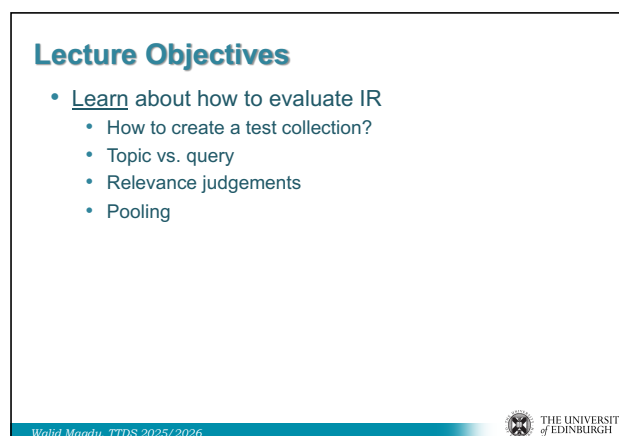
图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

Cranfield paradigm 回顾 → test collection 的来源：web logs 或 evaluation campaigns → TREC tracks 与 task → topic 不等于 query：description 与 narrative 定义相关性 → relevance judgments 的成本问题 → pooling 用多个系统的 top results 构造可评估候选集 → Cohen's kappa 衡量标注者一致性 → web A/B testing 与 offline evaluation 互补 → significance testing 判断分数提升是否可靠

10/14/25



1



2

1

这节课一句话

这节课一句话：TTDS 10 不是继续发明新指标，而是解释这些指标背后的“标准答案”从哪里来，以及平均分差异何时才算真实改进。

1. 1. Test collection 从哪里来

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Where Do Test Collections Come From?

- For web search, companies apply their own studies to assess the performance of their search engine.
- Web-search performance is monitored by:
 - Traffic
 - User clicks and session logs
 - Labelling results for selected users' queries
- For other search tasks:
 - Someone goes out and builds them (expensive)
 - As the by-product of large scale evaluations
- IR Evaluation Campaigns are created for this reason

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IR Evaluation Campaigns

- IR test collections are provided for scientific communities to develop best IR methods
- Collections and queries are provided, relevance judgements are built during the campaign
- TREC = Text REtrieval Conferences
 - Main IR eval campaign. Sponsored by NIST (US gov)
 - Series of annual evaluations, started in 1992
 - Organized into "tracks"
- Other evaluation campaigns
 - CLEF: European version (since 2000)
 - NTCIR: Asian version (since 1999)
 - FIRE: Indian version (since 2008)

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3

原 PPT 第 3 页

人话直觉

评价系统需要“考卷”和“答案”。Google 这类公司可以从真实用户行为里造题，学术社区则常靠大型评测活动一起造题。

精确定义 / 机制：Web search companies 可以监控 traffic、click/session logs，并对 selected queries 做人工标注；非 web search 任务通常没有现成大规模日志，所以需要 evaluation campaigns 构建 collection、topics 和 judgments。第 3 页强调 test collection 很贵，campaigns 的价值就是把成本集中投入，形成可复用资源。

考试问法：若问 why evaluation campaigns，答构建标准 test collections 成本高，需要社区共享任务、

数据和 judgments 来比较系统。

2. 2. TREC 与 track: 不是一个比赛, 而是一套任务生态

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TREC Task

- It is a task for search a set of documents of given genre and domain.
- TREC (or other IR eval campaigns) are formed of a set of tracks, each track has a set of search tasks.
- Example
 - TREC Medical track
 - TREC Legal track → CLEF-IP track → NTCIR patent mining track
 - TREC Microblog track
 - Different CLIR tracks in all campaigns

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TREC Collection

- 100's of collections were released in the different evaluation campaigns covering most of the domains in life
- A set of hundreds of thousands of docs
 - 1B in case of web search (TREC ClueWeb09)
- The typical format:

```
<DOC>
<DOCNO> 1234 </DOCNO>
<TEXT>
Multilines of plain text of the document
</TEXT>
</DOC>
```

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原 PPT 第 4 页

人话直觉

TREC 像 IR 的标准化考场; 不同 track 是不同科目的试卷, 比如 medical、legal、microblog、CLIR。

精确定义 / 机制: TREC 是由 NIST 赞助的主要 IR evaluation campaign。一个 track 围绕特定文档类型、领域或任务组织搜索任务, 提供 collection、topics、judgments 和评估协议。它的核心价值不是让某个系统赢一次, 而是让不同方法在同一任务定义下可比较、可复现、可累积。

考试问法: 若问 TREC track 是什么, 答特定 domain/genre/task 的评测任务集合, 并给 Medical/Legal/Microblog/CLIR 例子。

3. 3. Topic vs Query: 考试最容易混淆的一对

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TREC Topic

- Query sets are provided for each collection. Generated by experts and is associated with additional details. It is called **Topics**, and contains:
 - Query: the query text
 - Description: description of what is meant by the query
 - Narrative: what should be considered relevant

```
<num>189</num>
<title>Health and Computer Terminals</title>
<desc>Is it hazardous to the health of individuals to work with computer terminals on a daily basis?</desc>
<narr>Relevant documents would contain any information that expands on any physical disorder/problems that may be associated with the daily working with computer terminals. Such things as carpal tunnel, cataracts, and fatigue have been said to be associated, but how widespread are these or other problems and what is being done to alleviate any health problems</narr>
```

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Relevance Judgements

- For each topic, set of relevant docs is required to be known for an effective evaluation!
- Exhaustive assessment** is usually impractical
 - TREC usually has 50 topics
 - Collection usually has >1 million documents
- Random sampling** won't work
 - If relevant docs are rare, none may be found!
- IR systems** can help focus the sample (**Pooling**)
 - Each system finds some relevant documents
 - Different systems find different relevant documents
 - Together, enough systems will find most of them
 - Leverages cooperative evaluations

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原 PPT 第 5 页

人话直觉

query 是用户打在搜索框里的几个字；topic 是给 assessor 的完整案件说明书，告诉他什么算相关、什么不算。

精确定义 / 机制：TREC topic 通常包含 num、title/query、description 和 narrative。title 是短 query；description 解释信息需求；narrative 明确 relevance criteria，减少 assessor 之间的理解差异。第 5 页的 Health and Computer Terminals 示例展示了同一短标题背后还需要细化的判定边界。

考试问法：若问 topic 为什么比 query 更适合评测，答它降低歧义，让人工 judgments 更一致、更可复现。

4. 4. Relevance judgments 为什么不能 exhaustive

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Pooling

1. Systems submit top **1000** documents per topic
2. Top **100** documents from each are judged
 - Single pool, duplicates removed, random ranking
 - Judged by the person who developed the topic
3. Treat unevaluated documents as irrelevant
4. Compute MAP (or others) down to **1000** documents
 - To make pooling work:
 - Large number of reasonable systems participating
 - Systems must not all "do the same thing"

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Pooling, does it work?

- Judgments can't possibly be exhaustive!
It doesn't matter: relative rankings of different systems remain the same!
Chris Buckley and Ellen M. Voorhees. (2004) Retrieval Evaluation with Incomplete Information. SIGIR 2004.
- This is only one person's opinion about relevance
It doesn't matter: relative rankings remain the same!
Ellen Voorhees. (1998) Variations in Relevance Judgments and the Measurement of Retrieval Effectiveness. SIGIR 1998.
- What about hits 101 to 1000?
It doesn't matter: relative rankings remain the same!
- We can't possibly use judgments to evaluate a system that didn't participate in the evaluation!
Actually, we can!
Justin Zobel. (1998) How Reliable Are the Results of Large-Scale Information Retrieval Experiments? SIGIR 1998.

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原 PPT 第 6 页

人话直觉

如果 50 个 topics、百万文档，每个 topic-document pair 都人工判断，就像让老师批五千万份卷子，现实上不可行。

精确定义 / 机制：Exhaustive assessment 的规模是 topics × documents，成本巨大；random sampling 又因为 relevant documents 稀少，可能抽不到足够相关样本。因此需要一种聪明的候选收集方法：只判断最可能被系统返回、也最可能影响排名指标的文档。这正是 pooling 的动机。

考试问法：若问为什么不用 random sampling，答 relevant docs sparse，随机抽样效率低且可能无法覆盖真正重要的相关文档。

5. 5. Pooling: 用多个系统帮 assessor 找候选答案

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原 PPT 第 6 页

人话直觉

Pooling 像让很多学生先交出他们觉得可能正确的答案，老师只批这些高频候选；不是批完整宇宙，但覆盖了最可能影响比较的区域。

精确定义 / 机制：典型流程是：系统每个 topic 提交 top 1000，取每个系统 top 100 进入 pool，去重并随机排序，由 topic creator 判断；未评估文档通常视为 irrelevant，再计算 MAP 等指标到 rank 1000。Pooling 要有效，需要大量 reasonable systems，而且系统不能全做同一件事，否则 pool 会偏向单一方法。

考试问法：若考 pooling，必须写步骤、未评估文档处理方式、为什么需要 many diverse systems，以及它不是 exhaustive 但在实践中可用。

6. 6. Cohen's kappa: 相关性判断也要测可靠性

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Who decides a doc is relevant or not?

- The same doc can be seen relevant by me, but not you
- Sometimes, it would be useful to have multiple judgements on relevance on the same document
- How to measure agreement among different assessors?
- Cohen's *kappa*

$$\kappa = \frac{P(A) - P(E)}{1 - P(E)}$$

$P(A)$ – proportion of time judges agree (*inter-annotator agreement*)
 $P(E)$ – what agreement would be by chance

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Cohen's kappa

- Two judges (J_1 & J_2) annotating 50 docs for relevance
- $P(A) = \frac{20+15}{50} = 0.7$
- $P(E) = P(J_1, J_2 | rel) + P(J_1, J_2 | irrel)$
 - $P(rel) = P(J_1 | rel) \cdot P(J_2 | rel) = \frac{20+10}{50} \cdot \frac{20+5}{50} = 0.6 \times 0.5 = 0.3$
 - $P(irrel) = P(J_1 | irrel) \cdot P(J_2 | irrel) = \frac{20}{50} \cdot \frac{25}{50} = 0.4 \times 0.5 = 0.2$
- $\kappa = \frac{P(A) - P(E)}{1 - P(E)}$
- $= \frac{0.7 - 0.5}{1 - 0.5} = \frac{0.2}{0.5} = 0.4$

		J_1	
		Relevant	Irrelevant
J_2	Relevant	20	5
	Irrelevant	10	15

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原 PPT 第 7 页

人话直觉

两个 assessor 同意很多次不一定厉害；如果数据极不平衡，他们可能靠碰巧也常同意。Kappa 要扣掉 chance agreement。

精确定义 / 机制：Cohen's kappa 衡量两个 judges 的 agreement 超过随机一致性的程度。 $P(A)$ 是 observed agreement, $P(E)$ 是 by chance 的 expected agreement。 $\kappa=1$ 表示完全一致, $\kappa=0$ 表示只达到随机水平, $\kappa<0$ 比随机还差；第 8 页给出经验阈值：>0.8 good, 0.67–0.8 fair, <0.67 评测基础可疑。

$$\kappa = \frac{P(A) - P(E)}{1 - P(E)}$$

考试问法：若问公式，解释每个项；若问 relevance judgments 的问题，答主观性 + multiple assessors + agreement measure。

7. 7. Web Search A/B Testing: 真实用户流量里的实验

10/14/25

Web Search Engines: A/B testing

- Purpose: Test a single innovation
- Prerequisite: You have a large search engine up & running.
- Have most users use old system
- Divert a small proportion of traffic (e.g., 1%) to the new system that includes the innovation
- Evaluate with an "automatic" measure, like clickthrough on first result
- Now we can directly see if the innovation does improve user satisfaction.
- Probably the evaluation methodology that large search engines trust most

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Is system B really better than A?

- Given the results from a number of queries, B achieved better score than A. How can we conclude that ranking algorithm B is really better than algorithm A?

Experiment 1

Query	System A	System B
1	0.20	0.40
2	0.21	0.41
3	0.22	0.42
4	0.19	0.39
5	0.17	0.37
6	0.20	0.40
7	0.21	0.41
Average	0.20	0.40

Experiment 2

Query	System A	System B
1	0.02	0.76
2	0.39	0.07
3	0.16	0.37
4	0.58	0.21
5	0.04	0.02
6	0.09	0.91
7	0.12	0.46
Average	0.20	0.40

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原 PPT 第 9 页

人话直觉

Offline evaluation 像模拟考；A/B testing 像把新菜单给 1% 真顾客试吃，直接看他们会不会更满意。

精确定义 / 机制： A/B testing 的目的通常是测试单个 innovation。大多数用户继续用 old system，少量流量导向 new system，用 clickthrough on first result 等 automatic measures 观察用户满意度是否改善。它需要已有大型 search engine 和足够流量，因此是大公司最信任但不是所有任务都能使用的方法。

考试问法：若问 web search 公司如何评估，答 traffic、logs、selected query labeling、A/B testing；若问为什么 recall 难测，说明 web 全体 relevant docs 不可知。

8. 8. Significance Testing: 平均分高不是结案陈词

10/14/25

Significance Test

- It is not enough to show that system B achieves better average score than system A
 - Significance test is essential
- Two-tailed t-test is highly accepted, with $\alpha=0.05$
 - Sometimes it is required to use others
 - Wilcoxon test: does not assume normal distribution
- Meaning of significance test for IR system
 - When a user uses system B that is significantly better than system A, he/she will feel the difference in performance
 - If system B is better than A but not significantly, the user won't notice a difference between the two systems

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Now, is system B better than A?

Experiment 1			Experiment 2		
Query	System A	System B	Query	System A	System B
1	0.20	0.40	1	0.02	0.76
2	0.21	0.41	2	0.39	0.07
3	0.22	0.42	3	0.16	0.37
4	0.19	0.39	4	0.58	0.21
5	0.17	0.37	5	0.04	0.02
6	0.20	0.40	6	0.09	0.91
7	0.21	0.41	7	0.12	0.46
Average	0.20	0.40	Average	0.20	0.40

t-test p-value = 0
B is statistically significantly better than A

t-test p-value = 0.306
B and A are statistically indistinguishable

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原 PPT 第 12 页

人话直觉

系统 B 平均 MAP 高一点，可能是真强，也可能只是这组 queries 运气好。显著性检验是在问：这个差异像不像随机波动？

精确定义 / 机制：Null hypothesis 通常是假设两个系统没有真实差异，observed differences 只是随机波动。Significance test 试图拒绝 null hypothesis；test power 是正确拒绝 null 的概率，增加 query 数量通常提高 power。IR 中常用 two-tailed t-test， $\alpha=0.05$ ；如果不能假设 per-query differences 近似正态，可以考虑 Wilcoxon。

考试问法：若问为什么平均分更高还不够，答需要统计显著性；若问 t-test 假设，答 per-query effectiveness differences 来自 normal distribution 且均值差为 0 是 null。

考试速记版

- Test collection = documents + topics + relevance judgments + measure。

- Topic = query/title + description + narrative; query 只是短文本。
- Pooling = 多系统 top results 合并去重后人工判断。
- Pooling 需要 many reasonable and diverse systems。
- Unevaluated docs 通常当作 irrelevant。
- Cohen's kappa 扣除 chance agreement 后衡量 assessor agreement。
- A/B testing 适合已有大规模真实流量的 web search。
- Significance test 用来判断平均分提升是否可能只是随机波动。

一段稳妥考场回答

A reusable IR test collection is normally built from a document collection, a set of topics, relevance judgments and an evaluation measure. In TREC-style evaluation, a topic is richer than a query because it includes a short title/query, a description of the information need, and a narrative specifying relevance criteria. Since exhaustive judging is too expensive and random sampling rarely finds enough relevant documents, pooling is used: multiple diverse systems submit top-ranked documents, the top results are merged and de-duplicated, assessors judge the pool, and unjudged documents are usually treated as non-relevant. Finally, a higher average score is not sufficient evidence that one system is better; a significance test, often a two-tailed t-test at $\alpha=0.05$ or a non-parametric alternative such as Wilcoxon, is needed to assess whether the improvement is likely to be real.